WASP-36b

R-0687

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_260125` · created 2026-07-27T20:16:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461065.9268 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.049
Transit depth (Rp/Rs)^2	2.78 +/- 1.64 [%]
Orbital Inclination (inc)	87.72 +/- 2.82
Airmass coefficient 1 (a1)	0.55 +/- 0.29
Airmass coefficient 2 (a2)	0.18 +/- 0.25
Scatter in the residuals of the lightcurve fit is	55.66 %
Best Comparison Star	#1 - [392, 242]
Optimal Aperture	0.99
Optimal Annulus	5.75
Transit Duration (day)	0.0933 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.049 vs 0.1368 (deviation 0.62σ)
Duration fit vs published	0.0933 d vs 0.0758 d (deviation 2.3σ)
Mid-time O–C	9.9 min
Depth SNR	3.4
Residual scatter	55.66 %

Light curve

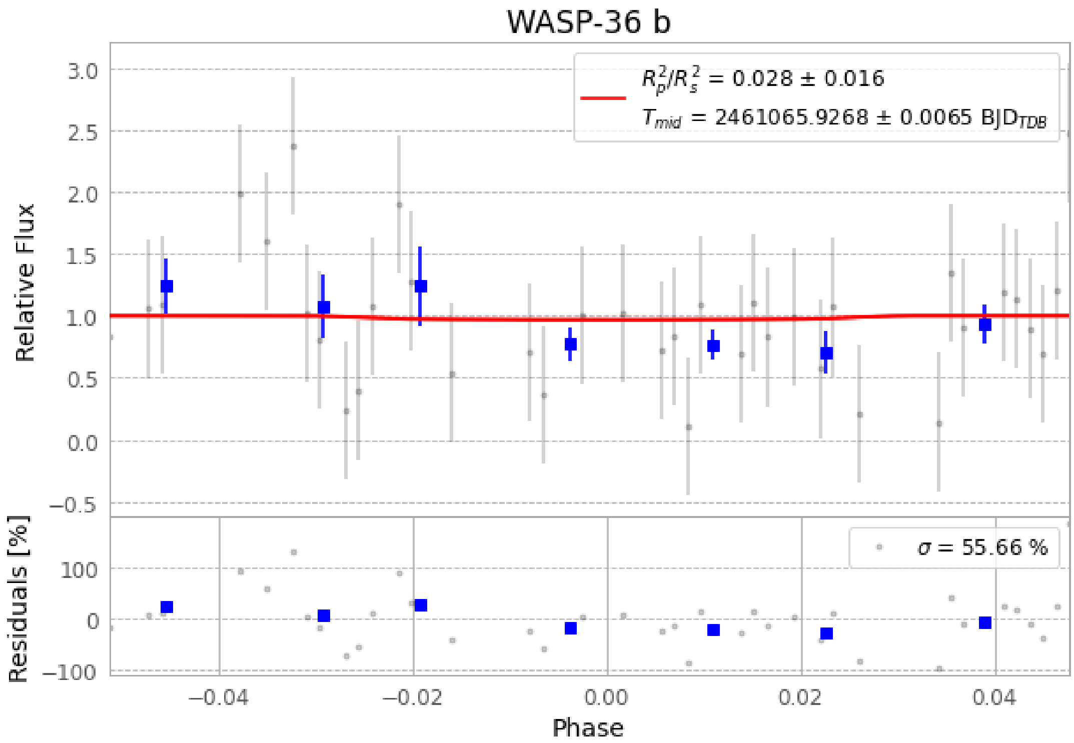

Figure 1 — FinalLightCurve_WASP-36 b_2026-01-25.png (SHA-256 283e099f6352fd72…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [512, 115], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2d29aa5574f6e9eb…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813`

Data provenance

77 FITS frames (51 MB), `WASP-36260125080315.FITS` → `WASP-36260125115115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-260125.log (d5c982686aff…), FinalLightCurve_WASP-36 b_2026-01-25.png (283e099f6352…), FinalLightCurve_WASP-36 b_2026-01-25.csv (2f1a6c4bc38d…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)